Parkinson's Disease

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Microbiome Signatures, Microbiome Medicine

Abstract. Explore Parkinson's disease causes and how the gut microbiome, metabolites, and metals may influence progression.

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Overview

Parkinson's disease (PD) is a neurodegenerative disorder primarily characterized by the degeneration of dopaminergic neurons in the nigrostriatal pathway, leading to progressive hypokinetic movements [1], and a range of non-motor symptoms, including gastrointestinal (GI) dysfunction [2]. However, Parkinson's disease involves complex interactions that extend beyond neuronal degeneration, with growing evidence highlighting the roles of the gut microbiome, metabolomic signals, and metallomics. The gut-brain axis is a crucial bidirectional communication pathway implicated in PD pathogenesis, where gut dysbiosis, inflammation, and alpha-synuclein aggregation are linked to disease progression. [2][3][4]Alterations in the gut microbiota, including reduced diversity and shifts in bacterial populations, contribute to increased intestinal permeability and neuroinflammation, potentially influencing alpha-synuclein misfolding and its spread from the gut to the brain via the vagus nerve. [5][6][7]

Causes

Parkinson's disease is increasingly understood as the clinical endpoint of converging pathogenic processes rather than a single linear pathway, with genetic susceptibilities intersecting with mitochondrial dysfunction, oxidative stress, alpha-synuclein misfolding, neuroinflammation, and environmental toxicant exposure. These mechanistic frameworks are tightly interconnected, since mitochondrial impairment and redox imbalance can promote protein aggregation and microglial activation, while gut-derived inflammatory signals and endotoxins amplify neuronal injury through the gut–brain axis. In parallel, dysregulated metal homeostasis and chronic exposure to pesticides and heavy metals further destabilize dopaminergic neurons by exacerbating oxidative stress, disrupting proteostasis, and impairing organellar function, particularly within the substantia nigra. Together, these processes create self-reinforcing cycles of bioenergetic failure, proteotoxic stress, and inflammatory signaling that progressively erode nigrostriatal integrity, driving the characteristic motor and non-motor manifestations of PD.

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Metal Homeostasis

Transition metals like iron, zinc, copper, and manganese are crucial for the enzymatic machinery of organisms, but their imbalance can foster pathogenic environments within the gastrointestinal tract.

Parkinson’s Disease

Parkinson’s disease is increasingly recognized as a systemic disorder involving coordinated disturbances across the gut–brain axis, rather than a condition confined to dopaminergic neurodegeneration alone. Converging evidence implicates gut dysbiosis, altered microbial metabolites, impaired intestinal barrier integrity, and metal dyshomeostasis as upstream drivers of neuroinflammation and alpha-synuclein pathology. These interconnected microbiome, metabolomic, and metallomic signals provide a mechanistic framework for understanding disease initiation, progression, and therapeutic targeting beyond the central nervous system.

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Short-chain Fatty Acids (SCFAs)

Short-chain fatty acids are microbially derived metabolites that regulate epithelial integrity, immune signaling, and microbial ecology. Their production patterns and mechanistic roles provide essential functional markers within microbiome signatures and support the interpretation of MBTIs, MMAs, and systems-level microbial shifts across clinical conditions.

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A microbiome signature is defined through a comprehensive approach that integrates differential abundance analysis across multiple datasets, followed by interpretation for biological relevance. This rigorous process ensures that the identified signature is both statistically robust and biologically meaningful. Associations that do not meet statistical significance or lack mechanistic relevance are excluded from the final signature.
Relative Abundance (Class)
A microbiome signature is defined through a comprehensive approach that integrates differential abundance analysis across multiple datasets, followed by interpretation for biological relevance. This rigorous process ensures that the identified signature is both statistically robust and biologically meaningful. Associations that do not meet statistical significance or lack mechanistic relevance are excluded from the final signature.
Relative Abundance (Family)
A microbiome signature is defined through a comprehensive approach that integrates differential abundance analysis across multiple datasets, followed by interpretation for biological relevance. This rigorous process ensures that the identified signature is both statistically robust and biologically meaningful. Associations that do not meet statistical significance or lack mechanistic relevance are excluded from the final signature.
Relative Abundance (Genus)
A microbiome signature is defined through a comprehensive approach that integrates differential abundance analysis across multiple datasets, followed by interpretation for biological relevance. This rigorous process ensures that the identified signature is both statistically robust and biologically meaningful. Associations that do not meet statistical significance or lack mechanistic relevance are excluded from the final signature.
Relative Abundance (Kingdom)
A microbiome signature is defined through a comprehensive approach that integrates differential abundance analysis across multiple datasets, followed by interpretation for biological relevance. This rigorous process ensures that the identified signature is both statistically robust and biologically meaningful. Associations that do not meet statistical significance or lack mechanistic relevance are excluded from the final signature.
Relative Abundance (Order)
A microbiome signature is defined through a comprehensive approach that integrates differential abundance analysis across multiple datasets, followed by interpretation for biological relevance. This rigorous process ensures that the identified signature is both statistically robust and biologically meaningful. Associations that do not meet statistical significance or lack mechanistic relevance are excluded from the final signature.
Relative Abundance (Phylum)
A microbiome signature is defined through a comprehensive approach that integrates differential abundance analysis across multiple datasets, followed by interpretation for biological relevance. This rigorous process ensures that the identified signature is both statistically robust and biologically meaningful. Associations that do not meet statistical significance or lack mechanistic relevance are excluded from the final signature.
Relative Abundance (Species)

▶ **Studies**

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